

5. *Nereis virens* is a species of marine worm. A student wanted to investigate the effect of a change in solute concentration on these worms. The following method was used.
- Collect seawater and thirty worms.
 - Make a 10% solution of seawater by placing 1 dm³ of seawater and 9 dm³ of freshwater into a fish tank.
 - Record the mass of ten worms and place them into the fish tank of diluted seawater.
 - Every 15 minutes remove the worms, blot dry, re-weigh and record their mass.
 - Repeat this process, and after 75 minutes, return the worms to a container of undiluted seawater.
 - Repeat the experiment a further two times using different worms for each trial.
 - Calculate the mean percentage change in mass of the worms during the investigation.

The student’s results are shown in the table below.

Time / minutes	Mean percentage change in mass / %
0	0
15	15
30	18
45	20
60	22
75	22

(a) Explain why it was necessary for the student to calculate the percentage change in mass. [2]

.....

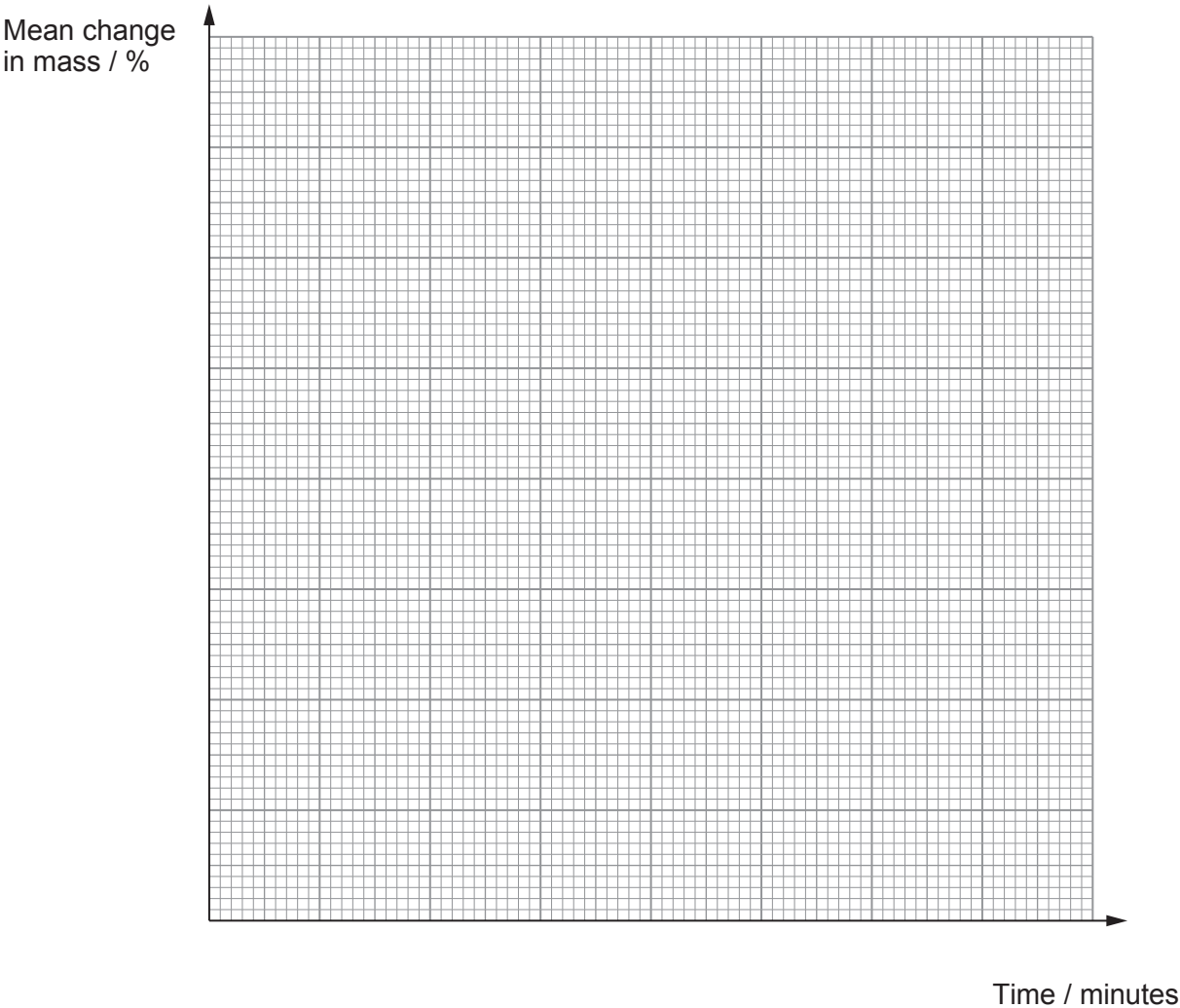
.....

.....



(b) (i) Plot the student's results shown in the table on the grid below.

[3]



(ii) Describe and explain the changes in mass of the worms during the experiment. [4]

.....

.....

.....

.....

.....

.....

.....

.....



Examiner
only

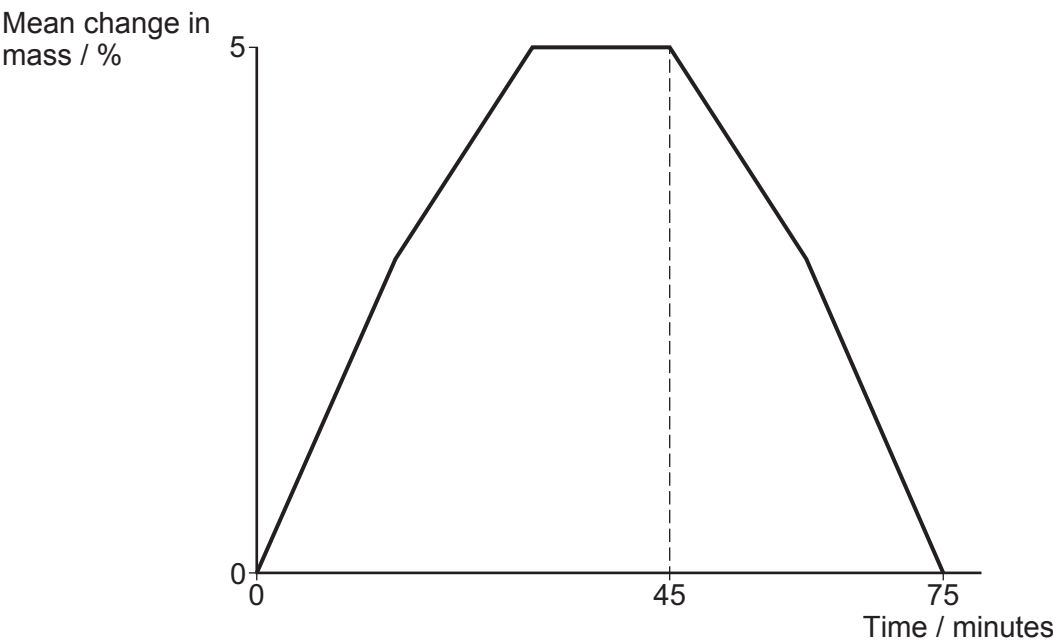
- (c) The student decided to repeat the experiment. She collected ten fresh worms, weighed them and placed them in a fish tank containing undiluted seawater. Explain why it was important for the student to carry out this second experiment. [2]

.....

.....

.....

- (d) A similar experiment was set up with a different species of marine worm, *Golfingia gouldi*. The results are shown below.



- (i) Describe how the results for *Golfingia* differ from those of *Nereis*. [2]

.....

.....

.....

.....



(ii) *Golfingia* can actively pump ions from its cells into the surrounding water. Explain the change in mass of *Golfingia* between 45 and 75 minutes. [2]

.....

.....

.....

.....

Examiner
only

15

